

Answer Key: Spring 2025 Operating Systems Midterm Exam

Operating Systems - Spring 2025

Multiple Choice Section (15 points total, 1 point each)

1. Which component of the Xinu operating system tracks process metadata?
A) The process table
2. What is the fundamental role of an operating system?
C) To manage hardware resources and provide services to applications
3. Which of the following best describes a lightweight process?
B) A thread that shares its address space with other threads
4. In the Producer-Consumer problem, what happens if the producer tries to add to a full buffer?
B) The producer goes to sleep until there is space
5. What scheduling algorithm would best address interactive systems where short tasks are delayed by long background processes?
A) Round Robin, because it enforces time-sharing by giving each process a fair slice of CPU time
6. Which of these is a primary cause of internal fragmentation?
A) Fixed-size memory allocation
7. What type of memory is used by the CPU for immediate processing and is the fastest?
B) Cache
8. What is an example of spatial multiplexing in an operating system?
B) Many virtual pages share a small number of physical page frames.
9. What information is saved during a context switch?
A) The process's CPU registers and program counter
10. What mechanism allows two or more processes to share a section of their address space while maintaining separate execution states?
B) Shared Memory Segments managed by the operating system
11. Which of the following is an example of a race condition?
B) Two threads accessing a shared variable without synchronization

12. What happens when a program accesses a page that is not in memory?
C) A page fault is triggered
13. A process control block (PCB) stores information about a process. Which of the following is NOT stored in a PCB?
C) System-wide file permission table
14. What is the primary function of Direct Memory Access (DMA) in an operating system?
B) To transfer data between peripherals and memory without CPU intervention
15. How does the Xinu readylist structure differ from a traditional scheduling queue?
B) It is a priority queue that dynamically reorders based on process priority

Short Answer / Essay Section (25 points total)

1. OS Kernel (5 points)

Describe the steps involved in handling a system call from a user process:

- The process executes a system call instruction (trap).
- The CPU switches to kernel mode and invokes the corresponding handler.
- The kernel verifies the system call parameters.
- The requested service (e.g., file read/write) is executed.
- The kernel returns the results to the user process.
- The CPU switches back to user mode and resumes execution.

2. Process Model (5 points)

Purpose of Process Table Entry Fields in Xinu:

1. prstate – Current state of the process (e.g., ready, running, sleeping).
2. prprio – Priority of the process for scheduling.
3. prstkptr – Stack pointer (location of the process's stack top).
4. prstkbase – Base address of the process's stack.
5. prstklen – Length of the stack.
6. prhasmsg – Flag indicating if a message is waiting.
7. prmsg – Message received by the process.

3. Scheduler (5 points)

Convoy Effect in Xinu Scheduler & Impact of Dynamic Quantum:

- Convoy Effect: If a long process enters the CPU first, shorter processes must wait, reducing system responsiveness.
- Impact of Dynamic Quantum:
 - Reduces context switches for high-priority processes.
 - Improves overall system throughput by balancing CPU allocation.

4. Virtual Addresses (5 points)

Translation Lookaside Buffer (TLB) in the CPU Memory Management Unit:

- Stored in TLB: Recently used page table entries (mappings between virtual and physical addresses).
- Why added at the hardware level?
 - Speeds up address translation.
 - Reduces page table lookup overhead.

5. Page Replacement Algorithm (5 points)

Explanation of Least Recently Used (LRU) Algorithm:

- Keeps track of pages based on recent use.
- When a page frame needs to be replaced, the least recently used page is evicted.

Simulating LRU Page Replacement Algorithm for:

Page reference sequence: 1, 3, 0, 3, 5, 6, 3, 0, 1, 4, 3, 0, 6 (4 frames available)

Step-by-step tracking:

1. 1 → Page fault (Load into Frame 1)
2. 3 → Page fault (Load into Frame 2)
3. 0 → Page fault (Load into Frame 3)
4. 3 → No fault (Already in memory)
5. 5 → Page fault (Load into Frame 4)
6. 6 → Page fault (Replace 1, least recently used)
7. 3 → No fault
8. 0 → No fault
9. 1 → Page fault (Replace 5)
10. 4 → Page fault (Replace 6)
11. 3 → No fault
12. 0 → No fault
13. 6 → Page fault (Replace 1)

Total Page Faults: 8